```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
Device: cuda
GPU: Tesla T4

Dataset:
Training images: 54
Testing images : 27

All training images annotation coverage:
  MA: 54/54 masks available; 0 missing
  HE: 53/54 masks available; 1 missing
  EX: 54/54 masks available; 0 missing
  SE: 26/54 masks available; 28 missing

Test images annotation coverage:
  MA: 27/27 masks available; 0 missing
  HE: 27/27 masks available; 0 missing
  EX: 27/27 masks available; 0 missing
  SE: 14/27 masks available; 13 missing

Warning: complete stratification is not possible. Using a seeded random split.

Final split:
Train: 46
Validation: 8
Test: 27

Training split annotation coverage:
  MA: 46/46 masks available; 0 missing
  HE: 45/46 masks available; 1 missing
  EX: 46/46 masks available; 0 missing
  SE: 22/46 masks available; 24 missing

Validation split annotation coverage:
  MA: 8/8 masks available; 0 missing
  HE: 8/8 masks available; 0 missing
  EX: 8/8 masks available; 0 missing
  SE: 4/8 masks available; 4 missing

Test split annotation coverage:
  MA: 27/27 masks available; 0 missing
  HE: 27/27 masks available; 0 missing
  EX: 27/27 masks available; 0 missing
  SE: 14/27 masks available; 13 missing
Downloading: "https://download.pytorch.org/models/resnet34-b627a593.pth" to /root/.cache/torch/hub/checkpoints/resnet34-b627a593.pth

100%|██████████| 83.3M/83.3M [00:00<00:00, 202MB/s]

Model initialized.
Trainable parameters: 24522660

Starting training...

Epoch  1/40:  100%
  90/90  [00:55<00:00,    6.05it/s,  loss=0.8047,  lr=5.44e-05]
Epoch 1: training loss = 0.84048

Evaluating 8 images...

100%
  8/8  [00:05<00:00,    1.60it/s]
Validation mean AUPR: 0.06221
MA: AUPR=0.0046984002346568575 | Dice=0.003007813164927567 | Threshold=0.25
HE: AUPR=0.030593468081111255 | Dice=0.017313180595193163 | Threshold=0.4
EX: AUPR=0.2060292824395732 | Dice=0.1315585804749447 | Threshold=0.6
SE: AUPR=0.007522077025720979 | Dice=0.00860310972182823 | Threshold=0.45
Saved best checkpoint: /content/drive/MyDrive/GlucoIris/lesion_model/unet_lesions_idrid.pt

Epoch  2/40:  100%
  90/90  [00:57<00:00,    5.81it/s,  loss=0.7614,  lr=1.57e-04]
Epoch 2: training loss = 0.77832

Evaluating 8 images...

100%
  8/8  [00:05<00:00,    1.60it/s]
Validation mean AUPR: 0.09378
MA: AUPR=0.00708495117385826 | Dice=0.0033369459278764003 | Threshold=0.35
HE: AUPR=0.03408807638802003 | Dice=0.016719425755618765 | Threshold=0.4
EX: AUPR=0.32601307447815214 | Dice=0.2259151281191909 | Threshold=0.55
SE: AUPR=0.007929201815418492 | Dice=0.007498178904614269 | Threshold=0.4
Saved best checkpoint: /content/drive/MyDrive/GlucoIris/lesion_model/unet_lesions_idrid.pt

Epoch  3/40:  100%
  90/90  [00:54<00:00,    5.69it/s,  loss=0.7299,  lr=2.58e-04]
Epoch 3: training loss = 0.74208

Epoch  4/40:  100%
  90/90  [00:56<00:00,    4.75it/s,  loss=0.6741,  lr=3.00e-04]
Epoch 4: training loss = 0.70087

Evaluating 8 images...

100%
  8/8  [00:05<00:00,    1.61it/s]
Validation mean AUPR: 0.08298
MA: AUPR=0.009635117071197491 | Dice=0.004482244994689421 | Threshold=0.25
HE: AUPR=0.06252587751290556 | Dice=0.04476037311032198 | Threshold=0.35
```

```
EX: AUPR=0.2545242332426375 | Dice=0.1707607820762267 | Threshold=0.45
SE: AUPR=0.005245266995042931 | Dice=0.005529327759637824 | Threshold=0.3

Epoch  5/40:  100%
 90/90  [00:52<00:00,    3.83it/s,  loss=0.6490,  lr=2.99e-04]
Epoch 5: training loss = 0.65993

Epoch  6/40:  100%
 90/90  [00:55<00:00,    5.48it/s,  loss=0.6073,  lr=2.98e-04]
Epoch 6: training loss = 0.62920

Evaluating 8 images...

100%
 8/8  [00:05<00:00,    1.61it/s]
Validation mean AUPR: 0.16710
MA: AUPR=0.014788441268778528 | Dice=0.009840324873926224 | Threshold=0.25
HE: AUPR=0.2012990819193857 | Dice=0.14024390243901552 | Threshold=0.55
EX: AUPR=0.4412805409899805 | Dice=0.24583339781185137 | Threshold=0.35
SE: AUPR=0.011043980127658561 | Dice=0.00832419743695017 | Threshold=0.2
Saved best checkpoint: /content/drive/MyDrive/GlucoIris/lesion_model/unet_lesions_idrid.pt

Epoch  7/40:  100%
 90/90  [00:53<00:00,    4.94it/s,  loss=0.6134,  lr=2.95e-04]
Epoch 7: training loss = 0.60789

Epoch  8/40:  100%
 90/90  [00:55<00:00,    4.97it/s,  loss=0.5776,  lr=2.91e-04]
Epoch 8: training loss = 0.59139

Evaluating 8 images...

100%
 8/8  [00:05<00:00,    1.61it/s]
Validation mean AUPR: 0.13527
MA: AUPR=0.010149797102029163 | Dice=0.006822310178014615 | Threshold=0.15
HE: AUPR=0.20069087008319975 | Dice=0.17159420289854047 | Threshold=0.45
EX: AUPR=0.3165174698191505 | Dice=0.2003082307365927 | Threshold=0.5
SE: AUPR=0.013710643605844142 | Dice=0.025453894644734135 | Threshold=0.2

Epoch  9/40:  100%
 90/90  [00:51<00:00,    4.85it/s,  loss=0.5637,  lr=2.86e-04]
Epoch 9: training loss = 0.58027

Epoch  10/40:  100%
 90/90  [00:55<00:00,    5.53it/s,  loss=0.5648,  lr=2.80e-04]
Epoch 10: training loss = 0.56897

Evaluating 8 images...

100%
 8/8  [00:05<00:00,    1.61it/s]
Validation mean AUPR: 0.14115
MA: AUPR=0.012028066493416444 | Dice=0.006743613020359941 | Threshold=0.1
HE: AUPR=0.1791981084611855 | Dice=0.1509369676320144 | Threshold=0.55
EX: AUPR=0.36047577356844146 | Dice=0.20463275520893404 | Threshold=0.35
SE: AUPR=0.012917205765383823 | Dice=0.043771693019647236 | Threshold=0.15

Epoch  11/40:  100%
 90/90  [00:52<00:00,    5.58it/s,  loss=0.5483,  lr=2.73e-04]
Epoch 11: training loss = 0.56316

Epoch  12/40:  100%
 90/90  [00:57<00:00,    4.66it/s,  loss=0.5465,  lr=2.65e-04]
Epoch 12: training loss = 0.55620

Evaluating 8 images...

100%
 8/8  [00:05<00:00,    1.57it/s]
Validation mean AUPR: 0.14250
MA: AUPR=0.016279700186012692 | Dice=0.0014420650371321348 | Threshold=0.1
HE: AUPR=0.21467816483064092 | Dice=0.1656834966788854 | Threshold=0.4
EX: AUPR=0.3231618630084962 | Dice=0.20367142306208239 | Threshold=0.75
SE: AUPR=0.015869396488760353 | Dice=0.036500808334817135 | Threshold=0.1

Epoch  13/40:  100%
 90/90  [00:51<00:00,    4.01it/s,  loss=0.5510,  lr=2.56e-04]
Epoch 13: training loss = 0.55190

Epoch  14/40:  100%
 90/90  [00:57<00:00,    4.99it/s,  loss=0.5412,  lr=2.46e-04]
Epoch 14: training loss = 0.54663

Evaluating 8 images...

100%
 8/8  [00:05<00:00,    1.58it/s]
Validation mean AUPR: 0.14963
MA: AUPR=0.01214348042812951 | Dice=0.0031482893474670667 | Threshold=0.1
HE: AUPR=0.23512927109249226 | Dice=0.17166009044025404 | Threshold=0.6
EX: AUPR=0.3275265373848002 | Dice=0.2044541372674739 | Threshold=0.7
SE: AUPR=0.023701306149478817 | Dice=0.06905001531524568 | Threshold=0.2

Epoch  15/40:  100%
 90/90  [00:50<00:00,    4.85it/s,  loss=0.5264,  lr=2.36e-04]
Epoch 15: training loss = 0.54383
```

Epoch 16/40: 100%
90/90 [00:56<00:00, 4.80it/s, loss=0.5268, lr=2.25e-04]
Epoch 16: training loss = 0.53768

Evaluating 8 images...

100%
8/8 [00:05<00:00, 1.63it/s]
Validation mean AUPR: 0.15405
MA: AUPR=0.01428913532622952 | Dice=0.007364820229054584 | Threshold=0.1
HE: AUPR=0.13044228475371855 | Dice=0.08836717159298808 | Threshold=0.3
EX: AUPR=0.4428152698978196 | Dice=0.2437115173500025 | Threshold=0.25
SE: AUPR=0.0286544254859637 4 | Dice=0.05358351871553273 | Threshold=0.1

Epoch 17/40: 100%
90/90 [00:50<00:00, 4.22it/s, loss=0.5209, lr=2.13e-04]
Epoch 17: training loss = 0.53361

Epoch 18/40: 100%
90/90 [00:56<00:00, 4.46it/s, loss=0.5462, lr=2.01e-04]
Epoch 18: training loss = 0.53640

Evaluating 8 images...

100%
8/8 [00:05<00:00, 1.60it/s]
Validation mean AUPR: 0.13876
MA: AUPR=0.01379040972102891 | Dice=0.0 | Threshold=0.1
HE: AUPR=0.16781651476423953 | Dice=0.12770842061163573 | Threshold=0.55
EX: AUPR=0.3560462780530489 | Dice=0.20071693922386397 | Threshold=0.3
SE: AUPR=0.01739368412048467 | Dice=0.0 | Threshold=0.1

Epoch 19/40: 100%
90/90 [00:52<00:00, 5.61it/s, loss=0.5714, lr=1.89e-04]
Epoch 19: training loss = 0.53276

Epoch 20/40: 100%
90/90 [00:57<00:00, 5.45it/s, loss=0.5498, lr=1.76e-04]
Epoch 20: training loss = 0.52831

Evaluating 8 images...

100%
8/8 [00:05<00:00, 1.61it/s]
Validation mean AUPR: 0.15648
MA: AUPR=0.015306487825976614 | Dice=0.0039450613676198 33 | Threshold=0.15
HE: AUPR=0.19850205371804347 | Dice=0.14783214065989803 | Threshold=0.55
EX: AUPR=0.38552097170902994 | Dice=0.20666065506120823 | Threshold=0.15
SE: AUPR=0.026571331475115582 | Dice=0.025925053028507374 | Threshold=0.1

Epoch 21/40: 100%
90/90 [00:51<00:00, 5.32it/s, loss=0.5145, lr=1.63e-04]
Epoch 21: training loss = 0.52615

Epoch 22/40: 100%
90/90 [00:57<00:00, 5.25it/s, loss=0.5283, lr=1.50e-04]
Epoch 22: training loss = 0.52202

Evaluating 8 images...

100%
8/8 [00:05<00:00, 1.60it/s]
Validation mean AUPR: 0.14514
MA: AUPR=0.015942408469477434 | Dice=0.010521784877323038 | Threshold=0.1
HE: AUPR=0.22945500410931047 | Dice=0.17757843208178178 | Threshold=0.7
EX: AUPR=0.3019967022000215 | Dice=0.18197558727505703 | Threshold=0.1
SE: AUPR=0.033168896657401414 | Dice=0.08474072719791469 | Threshold=0.2

Epoch 23/40: 100%
90/90 [00:50<00:00, 4.18it/s, loss=0.5000, lr=1.37e-04]
Epoch 23: training loss = 0.51656

Epoch 24/40: 100%
90/90 [00:56<00:00, 5.48it/s, loss=0.4976, lr=1.24e-04]
Epoch 24: training loss = 0.52297

Evaluating 8 images...

100%
8/8 [00:05<00:00, 1.59it/s]
Validation mean AUPR: 0.16196
MA: AUPR=0.015533222452606556 | Dice=0.0 | Threshold=0.1
HE: AUPR=0.24477962147337612 | Dice=0.18191263870993293 | Threshold=0.55
EX: AUPR=0.36932294226462753 | Dice=0.21781135634663318 | Threshold=0.2
SE: AUPR=0.018217972020129116 | Dice=0.004008551576695168 | Threshold=0.1

Epoch 25/40: 100%
90/90 [00:50<00:00, 5.55it/s, loss=0.5463, lr=1.11e-04]
Epoch 25: training loss = 0.51318

Epoch 26/40: 100%
90/90 [00:55<00:00, 5.23it/s, loss=0.4559, lr=9.86e-05]
Epoch 26: training loss = 0.51242

Evaluating 8 images...

100%
8/8 [00:04<00:00, 1.64it/s]

Validation mean AUPR: 0.13985
MA: AUPR=0.016552102474591365 | Dice=0.0 | Threshold=0.1
HE: AUPR=0.19897840377842918 | Dice=0.15400992131132782 | Threshold=0.25
EX: AUPR=0.3135693954410539 | Dice=0.2017378076887541 | Threshold=0.45
SE: AUPR=0.030288881898904923 | Dice=0.0 | Threshold=0.1

Epoch 27/40: 100%
90/90 [00:53<00:00, 5.15it/s, loss=0.4868, lr=8.65e-05]
Epoch 27: training loss = 0.50675

Epoch 28/40: 100%
90/90 [00:57<00:00, 3.89it/s, loss=0.4673, lr=7.49e-05]
Epoch 28: training loss = 0.50978

Evaluating 8 images...

100%
8/8 [00:05<00:00, 1.60it/s]
Validation mean AUPR: 0.13620
MA: AUPR=0.015742669813611162 | Dice=0.0 | Threshold=0.1
HE: AUPR=0.19677233000014943 | Dice=0.15771104831221272 | Threshold=0.5
EX: AUPR=0.30919065074621577 | Dice=0.19705046255441414 | Threshold=0.6
SE: AUPR=0.023094664958515285 | Dice=0.0 | Threshold=0.1

Epoch 29/40: 100%
90/90 [00:52<00:00, 4.96it/s, loss=0.4593, lr=6.38e-05]
Epoch 29: training loss = 0.50253

Epoch 30/40: 100%
90/90 [00:56<00:00, 4.54it/s, loss=0.4790, lr=5.35e-05]
Epoch 30: training loss = 0.50118

Evaluating 8 images...

100%
8/8 [00:05<00:00, 1.59it/s]
Validation mean AUPR: 0.14740
MA: AUPR=0.018072860794299678 | Dice=0.0 | Threshold=0.1
HE: AUPR=0.21247780578707362 | Dice=0.16093032251824568 | Threshold=0.15
EX: AUPR=0.33992491415488224 | Dice=0.21865802114100627 | Threshold=0.25
SE: AUPR=0.019116628959007852 | Dice=0.035189518015893685 | Threshold=0.1

Epoch 31/40: 100%
90/90 [00:50<00:00, 5.35it/s, loss=0.5678, lr=4.38e-05]
Epoch 31: training loss = 0.49716

Epoch 32/40: 100%
90/90 [00:57<00:00, 5.39it/s, loss=0.5234, lr=3.50e-05]
Epoch 32: training loss = 0.49306

Evaluating 8 images...

100%
8/8 [00:05<00:00, 1.60it/s]
Validation mean AUPR: 0.13788
MA: AUPR=0.017197415487876577 | Dice=0.0 | Threshold=0.1
HE: AUPR=0.17976425992294365 | Dice=0.1288074924184722 | Threshold=0.15
EX: AUPR=0.32463144784077763 | Dice=0.20451294975575726 | Threshold=0.15
SE: AUPR=0.02991960154436461 | Dice=0.03646865620927772 | Threshold=0.1

Epoch 33/40: 100%
90/90 [00:51<00:00, 5.23it/s, loss=0.5142, lr=2.70e-05]
Epoch 33: training loss = 0.48998

Epoch 34/40: 100%
90/90 [00:58<00:00, 2.64it/s, loss=0.5476, lr=2.00e-05]
Epoch 34: training loss = 0.48947

Evaluating 8 images...

100%
8/8 [00:05<00:00, 1.57it/s]
Validation mean AUPR: 0.14284
MA: AUPR=0.02002192100435897 | Dice=0.0 | Threshold=0.1
HE: AUPR=0.1925106357305962 | Dice=0.14192634560905676 | Threshold=0.4
EX: AUPR=0.3376008891315315 | Dice=0.207460283614806 | Threshold=0.1
SE: AUPR=0.021225747506613935 | Dice=0.04752097252220449 | Threshold=0.1

Epoch 35/40: 100%
90/90 [00:53<00:00, 4.80it/s, loss=0.4552, lr=1.40e-05]
Epoch 35: training loss = 0.48717

Epoch 36/40: 100%
90/90 [00:58<00:00, 5.27it/s, loss=0.4446, lr=9.00e-06]
Epoch 36: training loss = 0.49409

Evaluating 8 images...

100%
8/8 [00:05<00:00, 1.58it/s]
Validation mean AUPR: 0.13995
MA: AUPR=0.019835881258049276 | Dice=0.0 | Threshold=0.1
HE: AUPR=0.19120314144347 07 | Dice=0.14217683906573 | Threshold=0.35
EX: AUPR=0.3257776228229077 | Dice=0.20397205691185163 | Threshold=0.1
SE: AUPR=0.02299307775553522 | Dice=0.011235955056173131 | Threshold=0.1

Epoch 37/40: 100%
90/90 [00:52<00:00, 5.10it/s, loss=0.4577, lr=5.07e-06]

```
Epoch 37: training loss = 0.48306

Epoch  38/40:  100%
 90/90  [00:58<00:00,    2.84it/s,  loss=0.5121,  lr=2.25e-06]
Epoch 38: training loss = 0.48924

Evaluating 8 images...

100%
 8/8  [00:05<00:00,    1.60it/s]
Validation mean AUPR: 0.14061
MA: AUPR=0.020123992013477414 | Dice=0.0 | Threshold=0.1
HE: AUPR=0.19497432039387336 | Dice=0.1466375530676343 | Threshold=0.15
EX: AUPR=0.3250408649008969 | Dice=0.19574790843062895 | Threshold=0.1
SE: AUPR=0.022303142224169983 | Dice=0.020634634994888948 | Threshold=0.1

Epoch  39/40:  100%
 90/90  [00:51<00:00,    4.57it/s,  loss=0.4980,  lr=5.59e-07]
Epoch 39: training loss = 0.48495

Epoch  40/40:  100%
 90/90  [00:54<00:00,    5.42it/s,  loss=0.5199,  lr=1.27e-09]
Epoch 40: training loss = 0.48961

Evaluating 8 images...

100%
 8/8  [00:05<00:00,    1.61it/s]
Validation mean AUPR: 0.13979
MA: AUPR=0.020008761567414 | Dice=0.0 | Threshold=0.1
HE: AUPR=0.1972062758191012 | Dice=0.14842179605584024 | Threshold=0.1
EX: AUPR=0.3164902027188924 | Dice=0.1922472752436378 | Threshold=0.1
SE: AUPR=0.02543493470749905 | Dice=0.008091446185455887 | Threshold=0.1

Training completed.
Best epoch: 6
Best validation mean AUPR: 0.16710301107645084
Best model: /content/drive/MyDrive/GlucoIris/lesion_model/unet_lesions_idrid.pt

Evaluating 8 images...

100%
 8/8  [00:05<00:00,    1.48it/s]
Frozen validation thresholds:
MA: 0.25
HE: 0.55
EX: 0.35
SE: 0.20

Evaluating 27 images...

100%
 27/27  [00:18<00:00,    1.30it/s]
======================================================================
FINAL IDRiD TEST RESULTS
======================================================================

MA — Microaneurysms
----------------------------------------------------------
accuracy             : 0.9758092738963936
precision            : 0.0013448633354735582
sensitivity          : 0.03181785534725568
specificity          : 0.976738656525421
dice                 : 0.002580648919608088
iou                  : 0.001291991548100418
TP                   : 886
TN                   : 27625789
FP                   : 657917
FN                   : 26960
aupr                 : 0.026416769764437645
threshold            : 0.25
annotated_pixels     : 28311552
positive_pixels      : 27846

HE — Haemorrhages
----------------------------------------------------------
accuracy             : 0.9695166481865773
precision            : 0.14737853952138136
sensitivity          : 0.388665027470875 9
specificity          : 0.9757746741155888
dice                 : 0.21371727090430004
iou                  : 0.1196435857679113
TP                   : 117289
TN                   : 27331232
FP                   : 678546
FN                   : 184485
aupr                 : 0.6088991625557432
threshold            : 0.55
annotated_pixels     : 28311552
positive_pixels      : 301774

EX — Hard Exudates
----------------------------------------------------------
accuracy             : 0.9349137058964478
precision            : 0.2470552230551655
sensitivity          : 0.18896422265140916
specificity          : 0.9716433112974391
dice                 : 0.21413998222458683
```

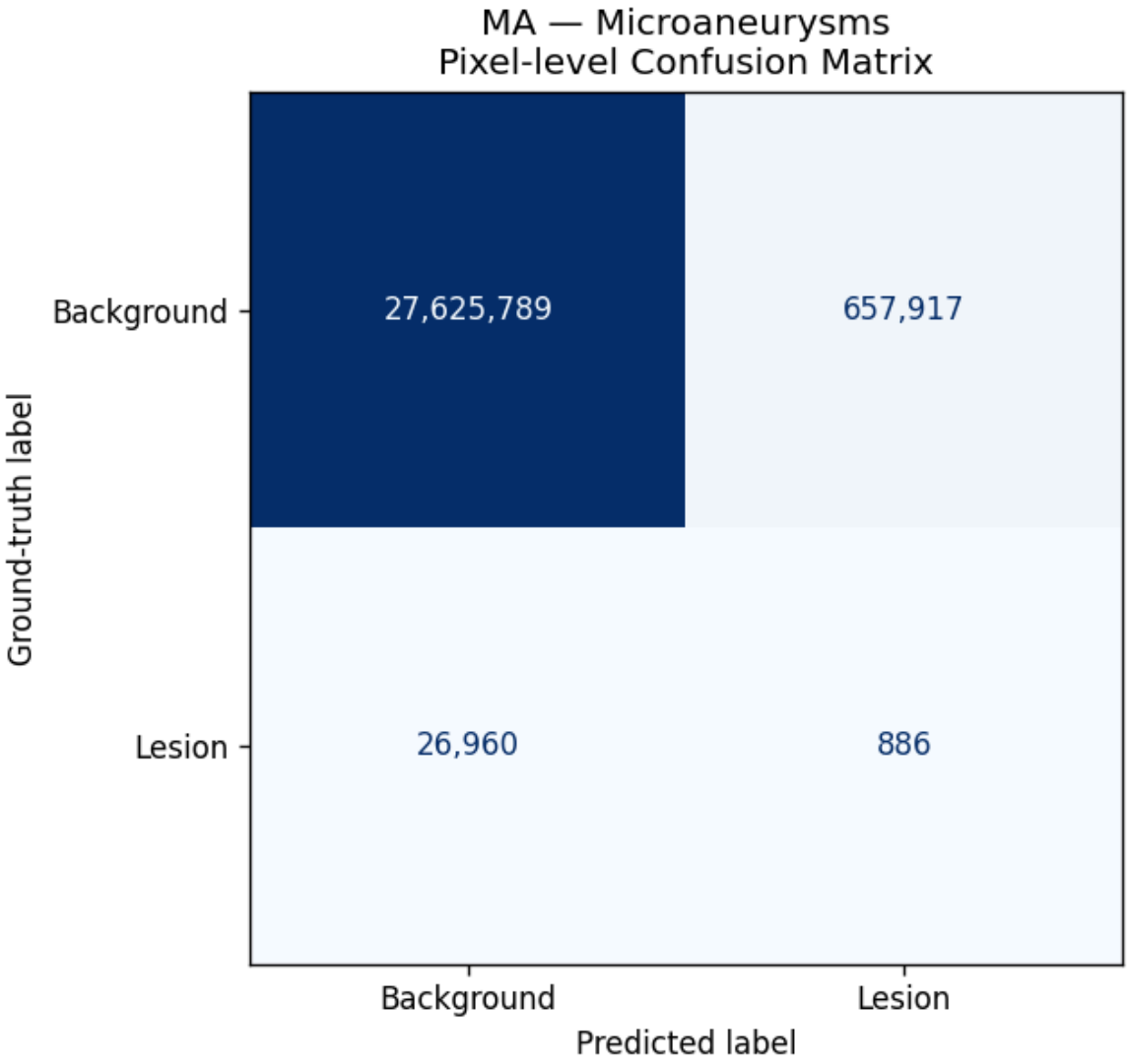
iou : 0.11990860431005892
TP : 251059
TN : 26217799
FP : 765147
FN : 1077547
aupr : 0.7760966913233662
threshold : 0.35
annotated_pixels : 28311552
positive_pixels : 1328606

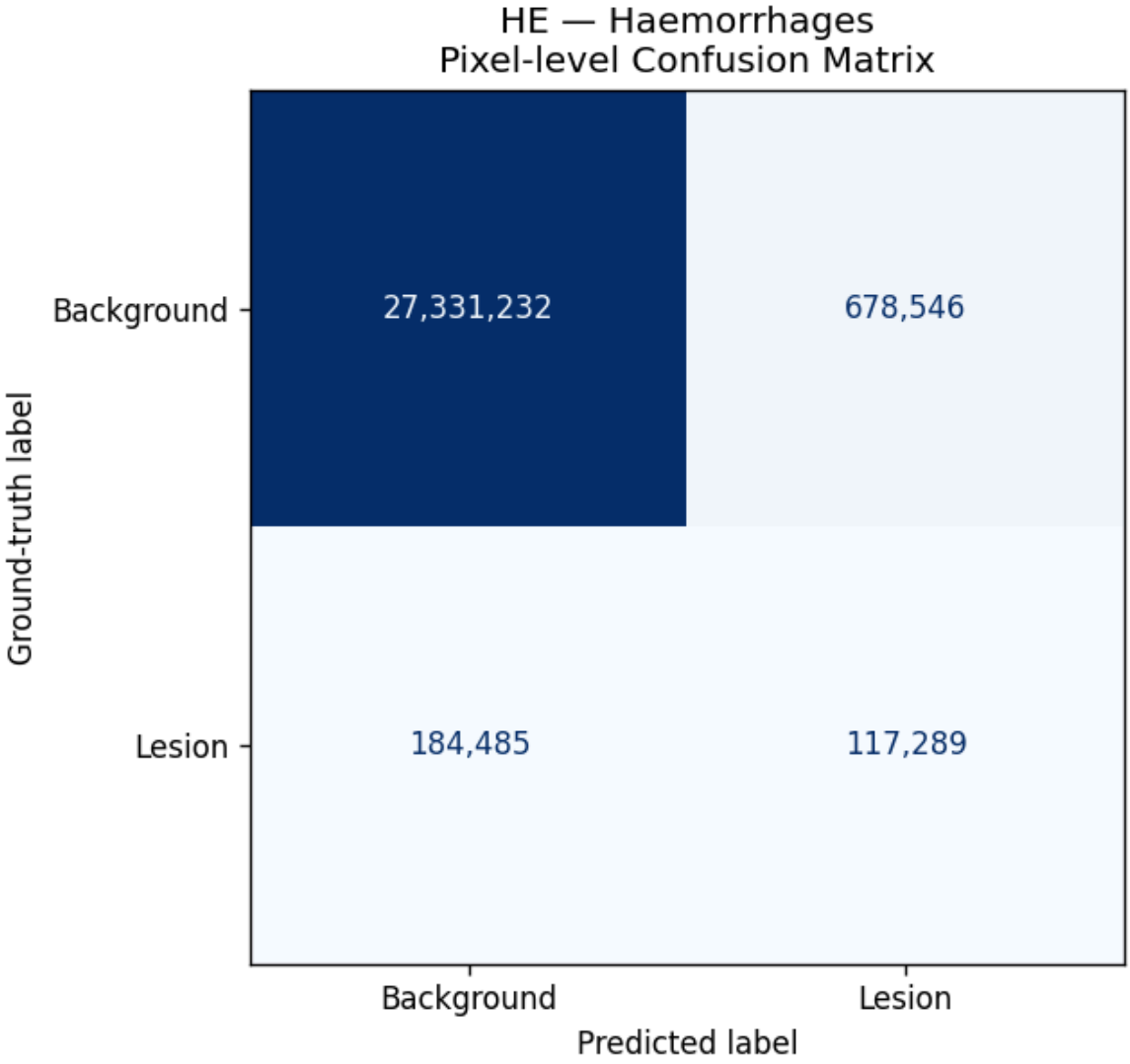
SE — Soft Exudates
---
accuracy : 0.7176554543631413
precision : 0.005918091032439776
sensitivity : 0.4793251089951946
specificity : 0.7184887601028507
dice : 0.011691826403775007
iou : 0.00588028886016606
TP : 24517
TN : 10510711
FP : 4118204
FN : 26632
aupr : 0.039088041775882876
threshold : 0.2
annotated_pixels : 14680064
positive_pixels : 51149

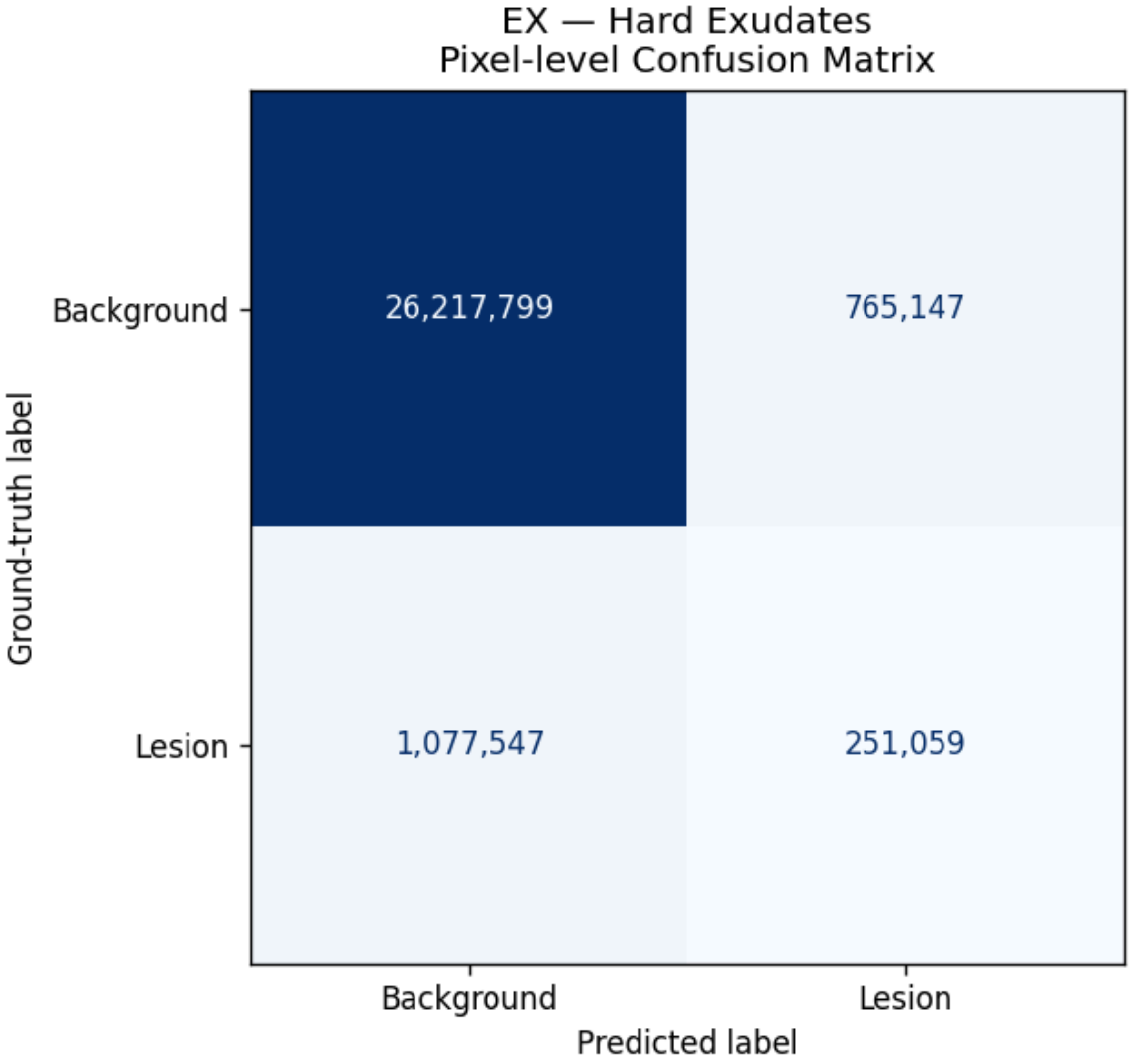
	Lesion	AUPR	Threshold	Dice	IoU	Accuracy	Precision	Sensitivity	Specificity	Annotated pixels	Positive pixels
0	MA	0.026417	0.25	0.002581	0.001292	0.975809	0.001345	0.031818	0.976739	28311552	27846
1	HE	0.608899	0.55	0.213717	0.119644	0.969517	0.147379	0.388665	0.975775	28311552	301774
2	EX	0.776097	0.35	0.214140	0.119909	0.934914	0.247055	0.188964	0.971643	28311552	1328606
3	SE	0.039088	0.20	0.011692	0.005880	0.717655	0.005918	0.479325	0.718489	14680064	51149

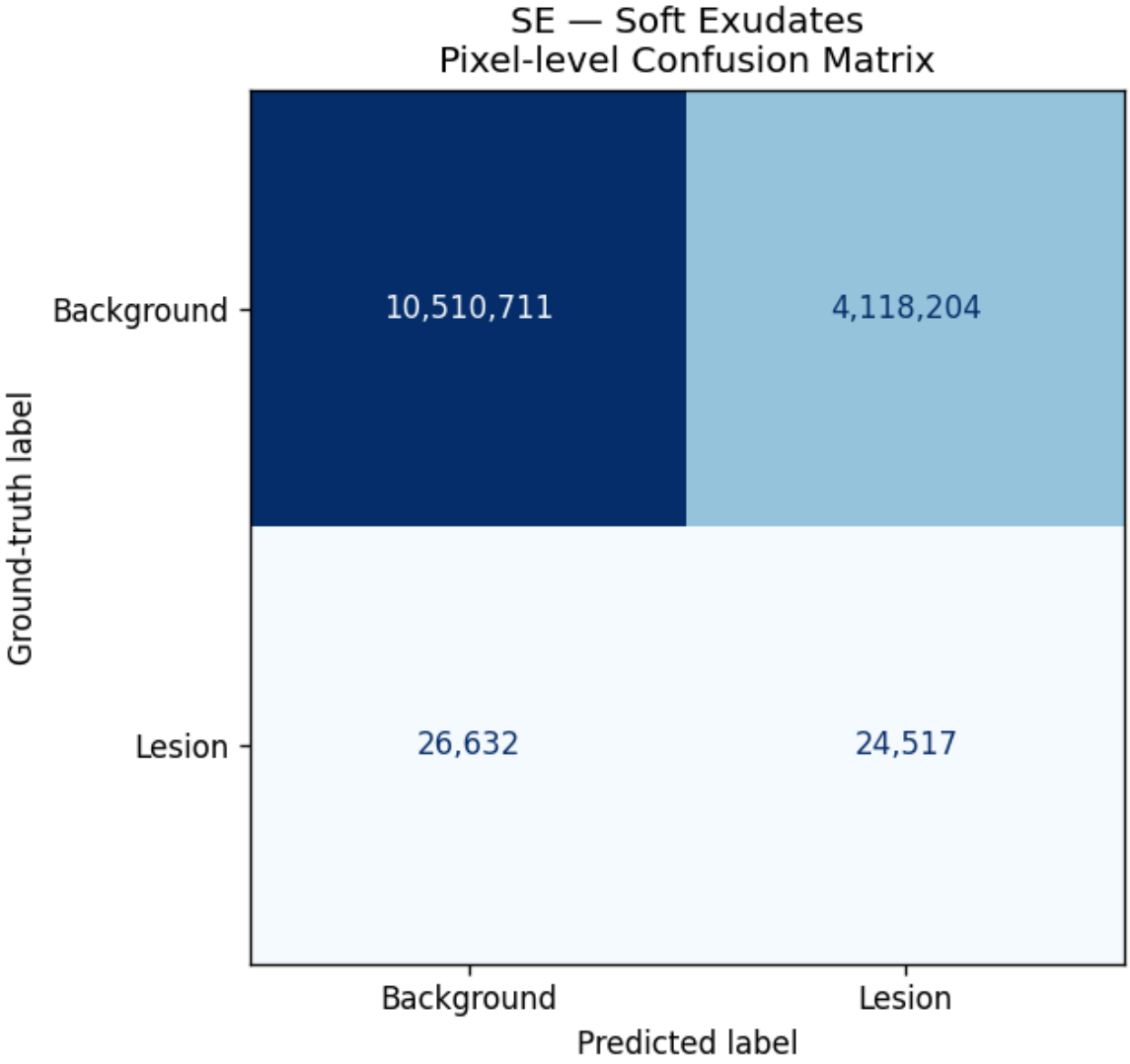
==========
SAVED OUTPUTS
==========
Best model : /content/drive/MyDrive/GlucoIris/lesion_model/unet_lesions_idrid.pt
Latest checkpoint: /content/drive/MyDrive/GlucoIris/lesion_model/unet_lesions_idrid_last.pt
Training history : /content/drive/MyDrive/GlucoIris/lesion_model/training_history.json
Thresholds : /content/drive/MyDrive/GlucoIris/lesion_model/lesion_thresholds.json
Final report : /content/drive/MyDrive/GlucoIris/lesion_model/lesions_idrid.json
Summary CSV : /content/drive/MyDrive/GlucoIris/lesion_model/lesion_test_summary.csv

GlucoIris lesion segmentation pipeline completed.

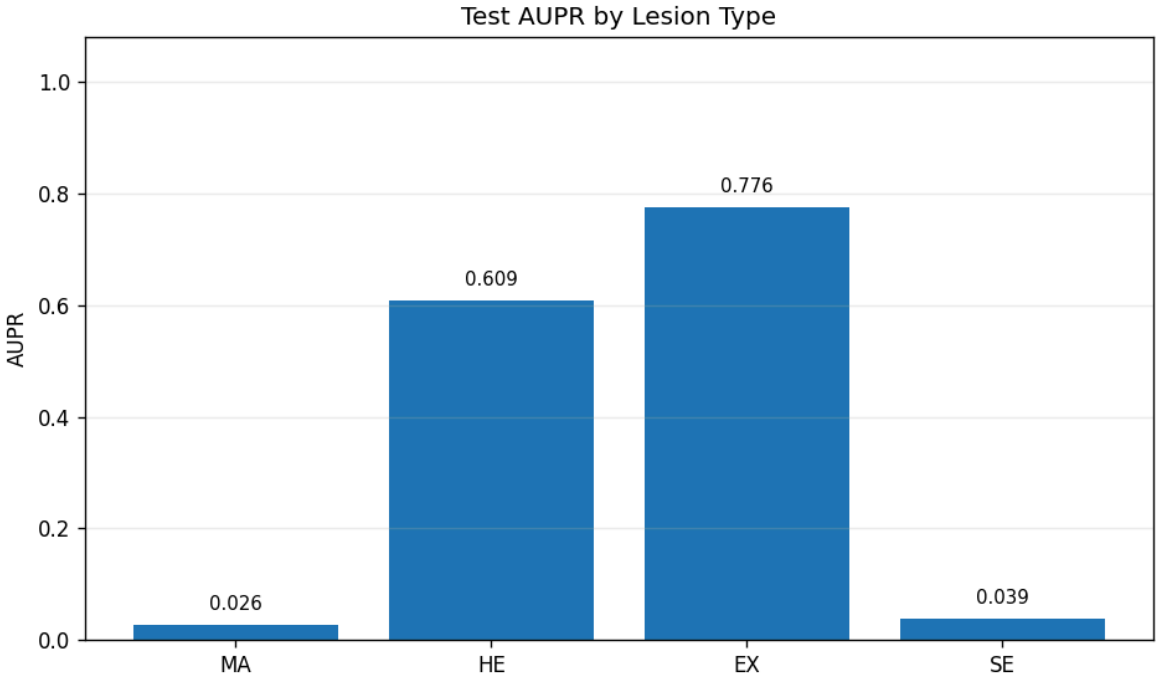

MA — Microaneurysms
Pixel-level Confusion Matrix
Background
Lesion
Ground-truth label
27,625,789
657,917
26,960
886
Background
Lesion
Predicted label

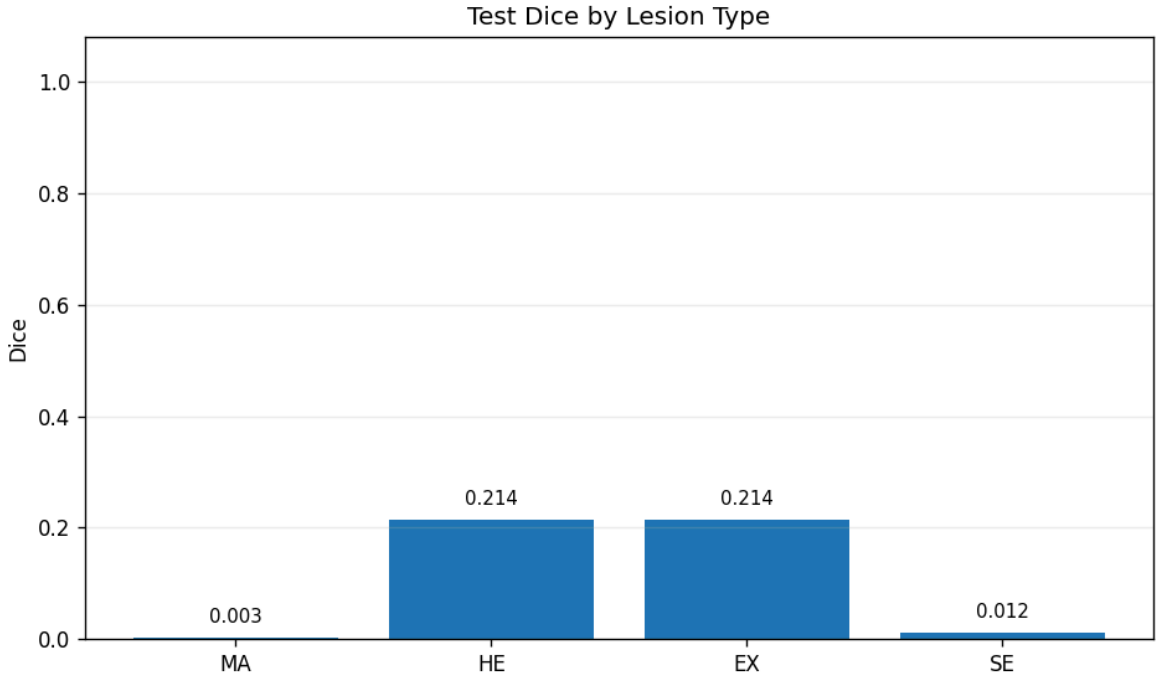

HE — Haemorrhages
Pixel-level Confusion Matrix
Background
Lesion
Ground-truth label
27,331,232
678,546
184,485
117,289
Background
Lesion
Predicted label

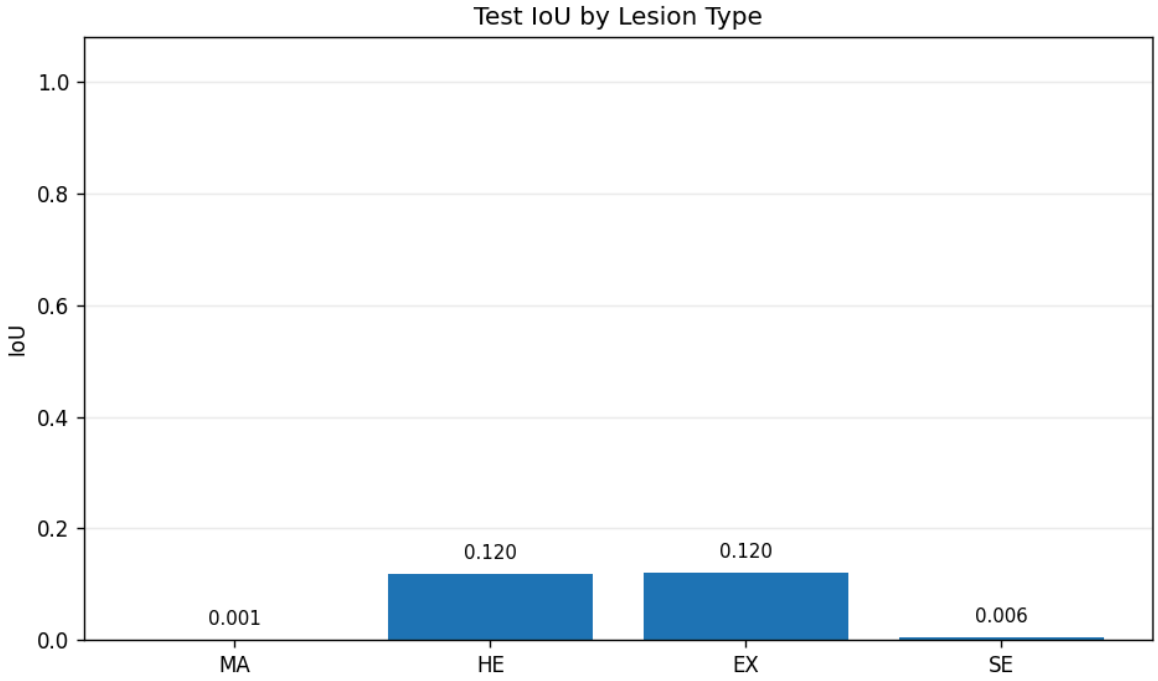

EX — Hard Exudates
Pixel-level Confusion Matrix
Background
Lesion
Ground-truth label
26,217,799
765,147
1,077,547
251,059
Background
Lesion
Predicted label

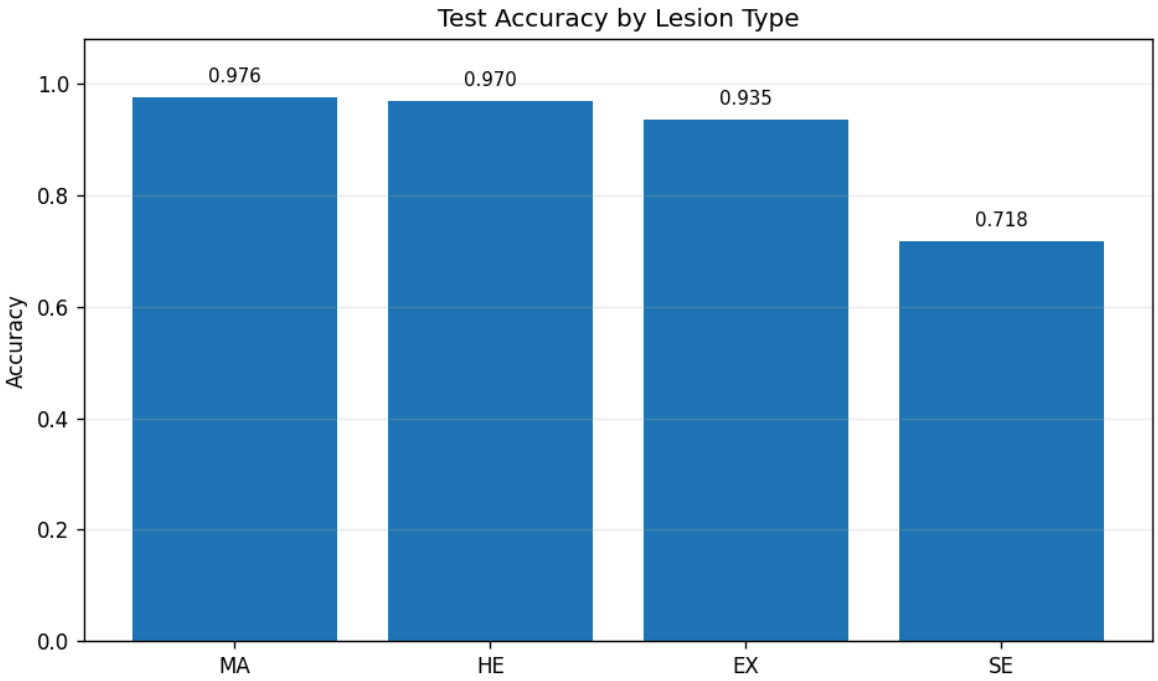

SE — Soft Exudates
Pixel-level Confusion Matrix
Background
Lesion
Ground-truth label
10,510,711
4,118,204
26,632
24,517
Background
Lesion
Predicted label

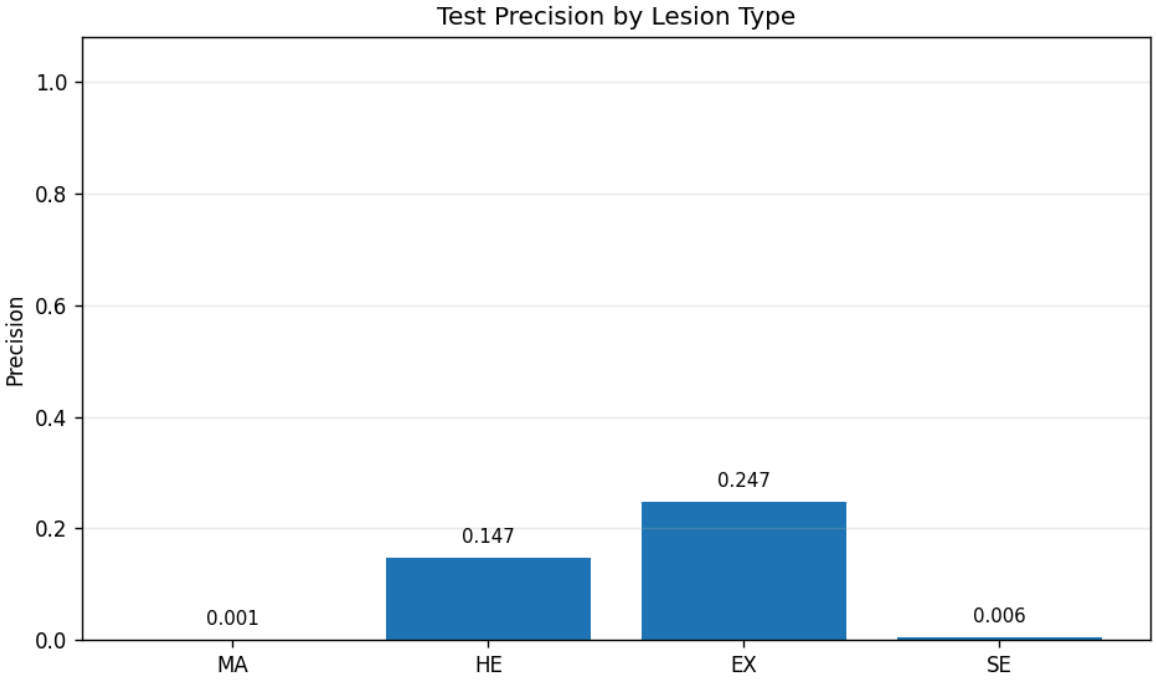

GlucoIris — IDRiD Test Confusion Matrices
MA — Microaneurysms
Background
Lesion
True label
27,625,789
657,917
26,960
886
Background
Lesion
Predicted label
HE — Haemorrhages
Background
Lesion
True label
27,331,232
678,546
184,485
117,289
Background
Lesion
Predicted label
EX — Hard Exudates
Background
Lesion
True label
26,217,799
765,147
1,077,547
251,059
Background
Lesion
Predicted label
SE — Soft Exudates
Background
Lesion
True label
10,510,711
4,118,204
26,632
24,517
Background
Lesion
Predicted label

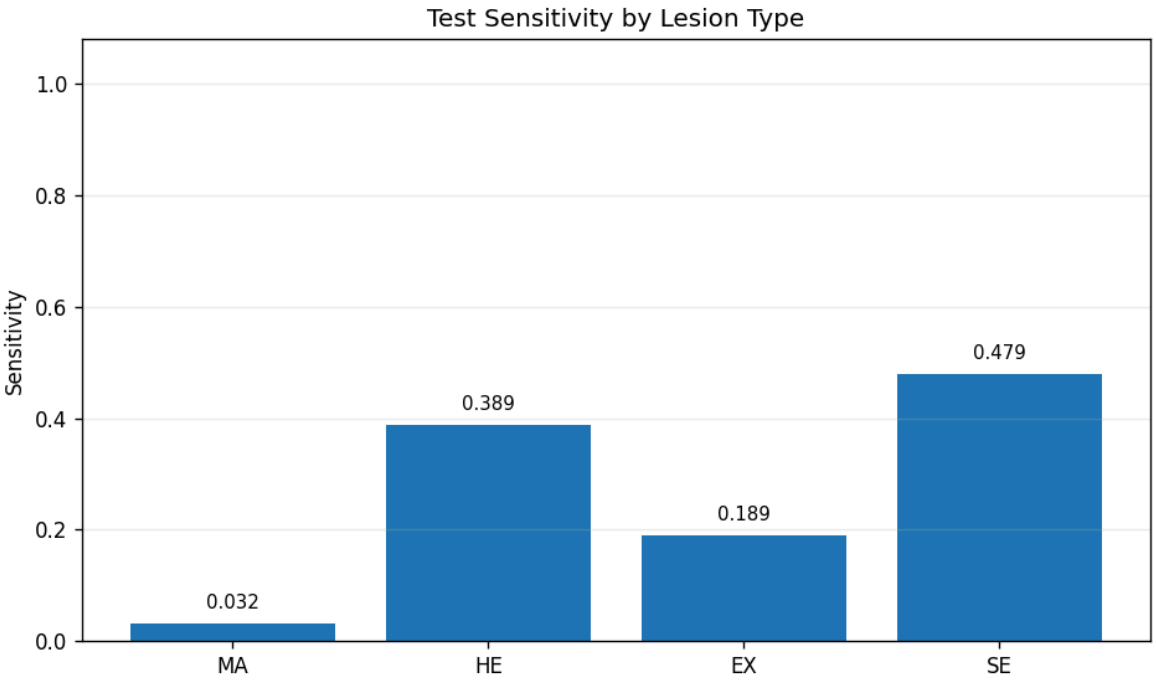

Test AUPR by Lesion Type
AUPR
1.0
0.8
0.6
0.4
0.2
0.0
0.026
0.609
0.776
0.039
MA
HE
EX
SE

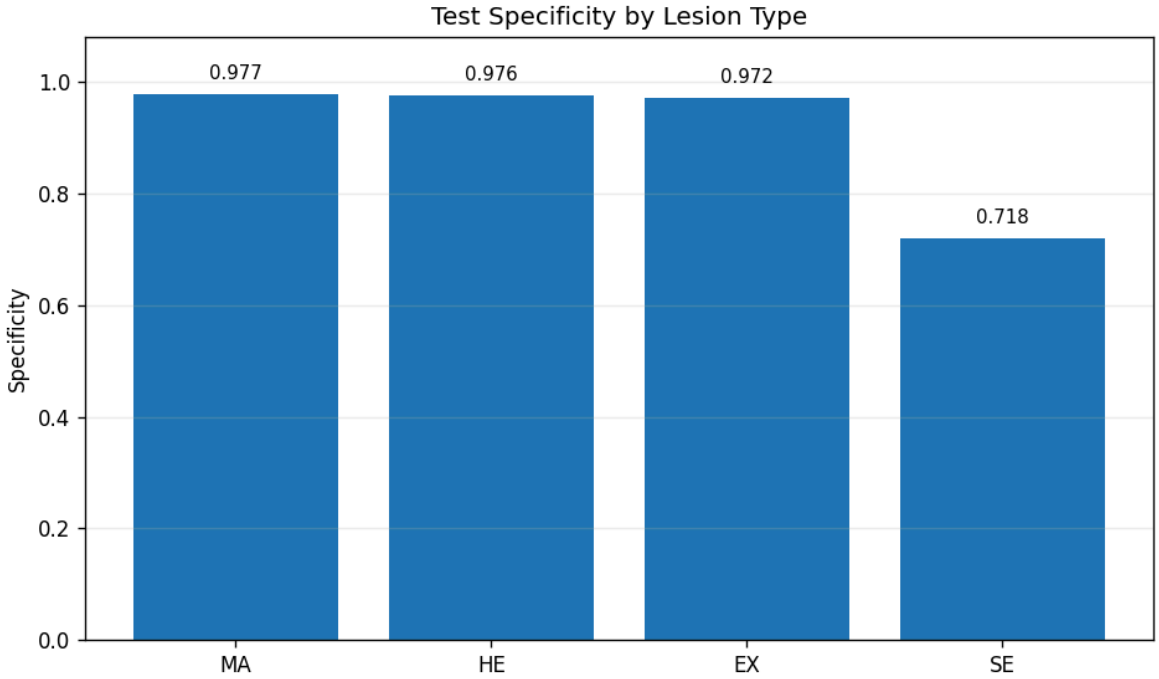

Test Dice by Lesion Type
Dice
1.0
0.8
0.6
0.4
0.2
0.0
0.003
0.214
0.214
0.012
MA
HE
EX
SE

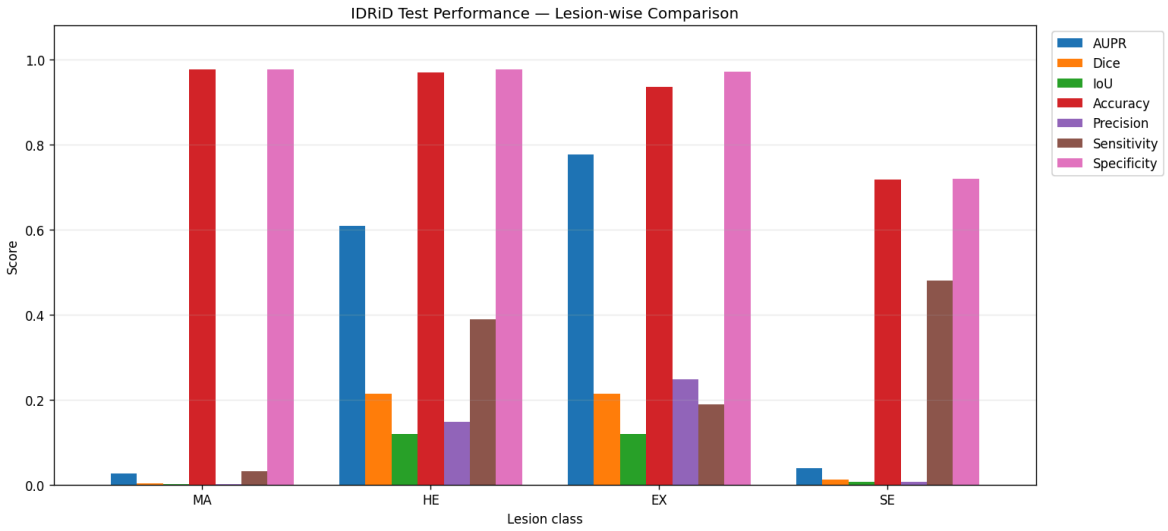

Test IoU by Lesion Type
IoU
1.0
0.8
0.6
0.4
0.2
0.0
0.001
0.120
0.120
0.006
MA
HE
EX
SE

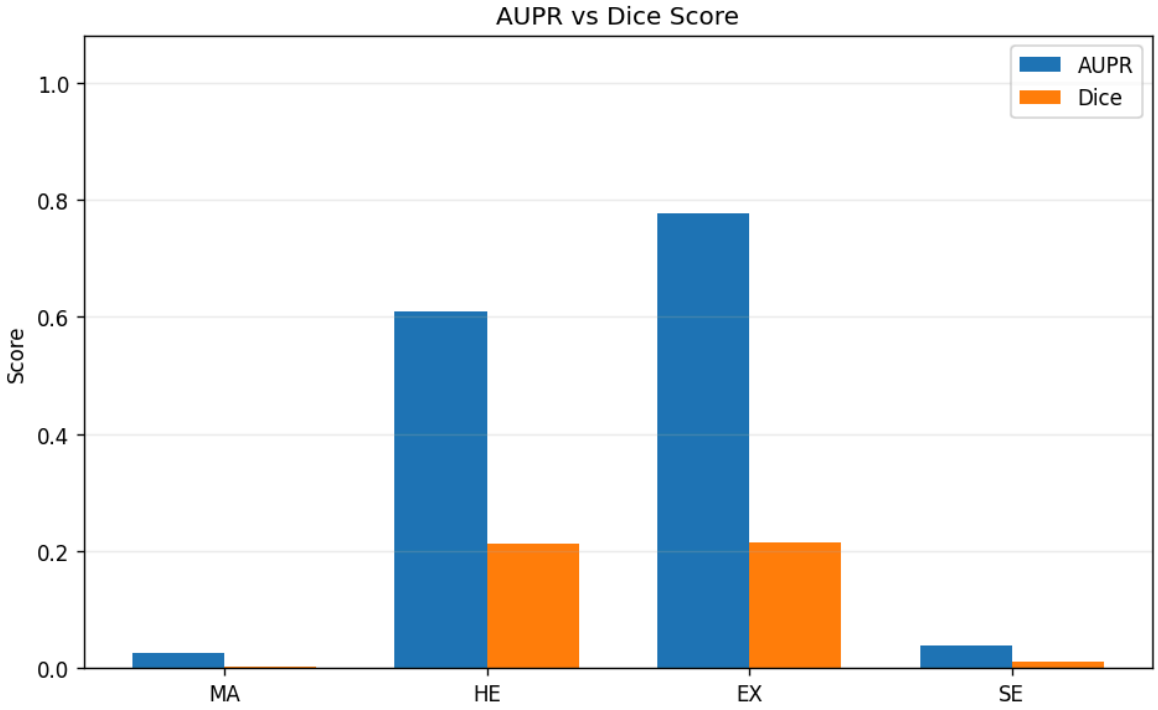

Test Accuracy by Lesion Type
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
0.976
0.970
0.935
0.718
MA
HE
EX
SE

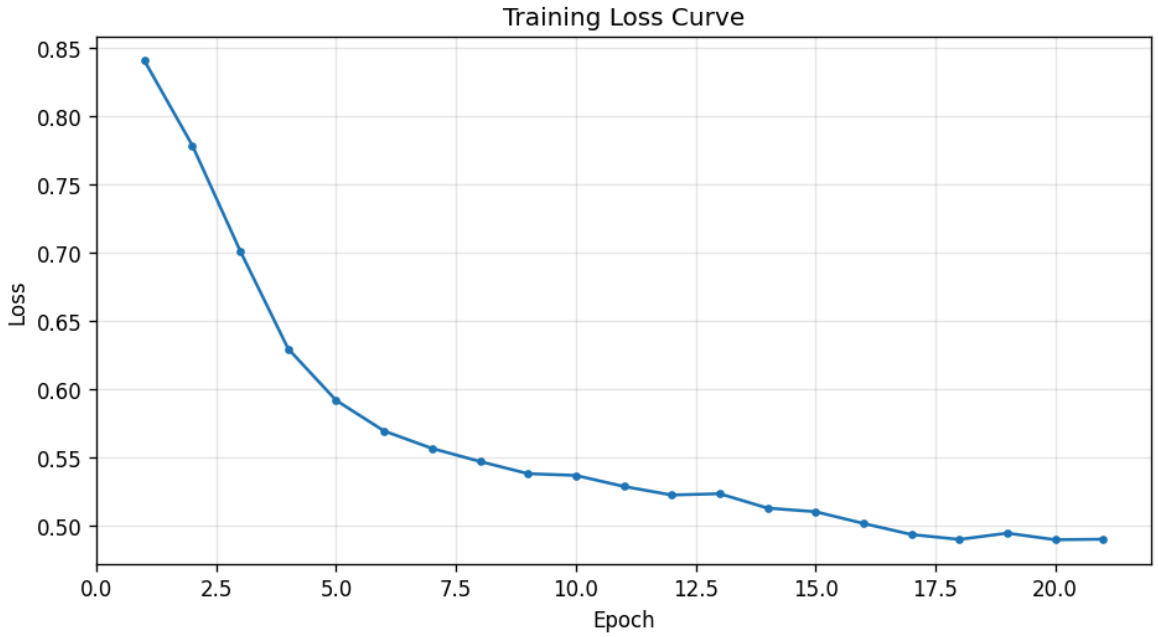

Test Precision by Lesion Type
Precision
1.0
0.8
0.6
0.4
0.2
0.0
0.001
0.147
0.247
0.006
MA
HE
EX
SE


Test Sensitivity by Lesion Type
Sensitivity
1.0
0.8
0.6
0.4
0.2
0.0
0.032
0.389
0.189
0.479
MA
HE
EX
SE


Test Specificity by Lesion Type
Specificity
1.0
0.8
0.6
0.4
0.2
0.0
0.977
0.976
0.972
0.718
MA
HE
EX
SE


IDRiD Test Performance — Lesion-wise Comparison
Score
1.0
0.8
0.6
0.4
0.2
0.0
MA
HE
EX
SE
Lesion class
AUPR
Dice
IoU
Accuracy
Precision
Sensitivity
Specificity





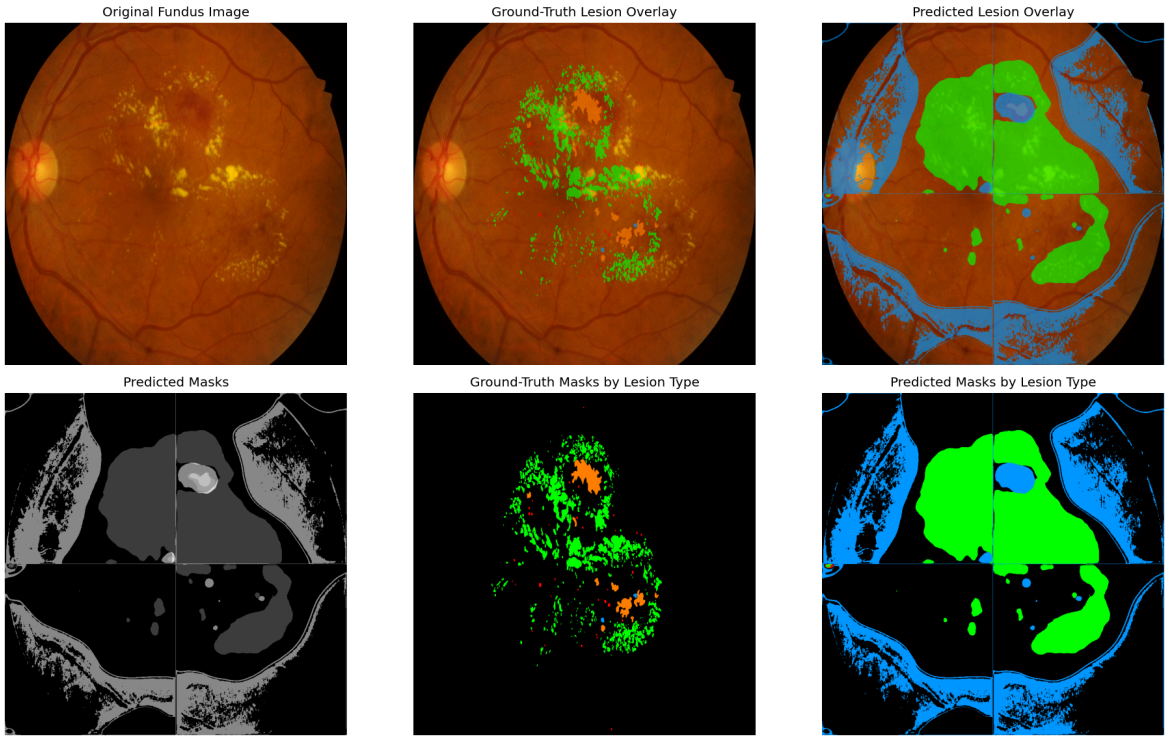
No usable values found for Loss.
No usable values found for Mean AUPR.

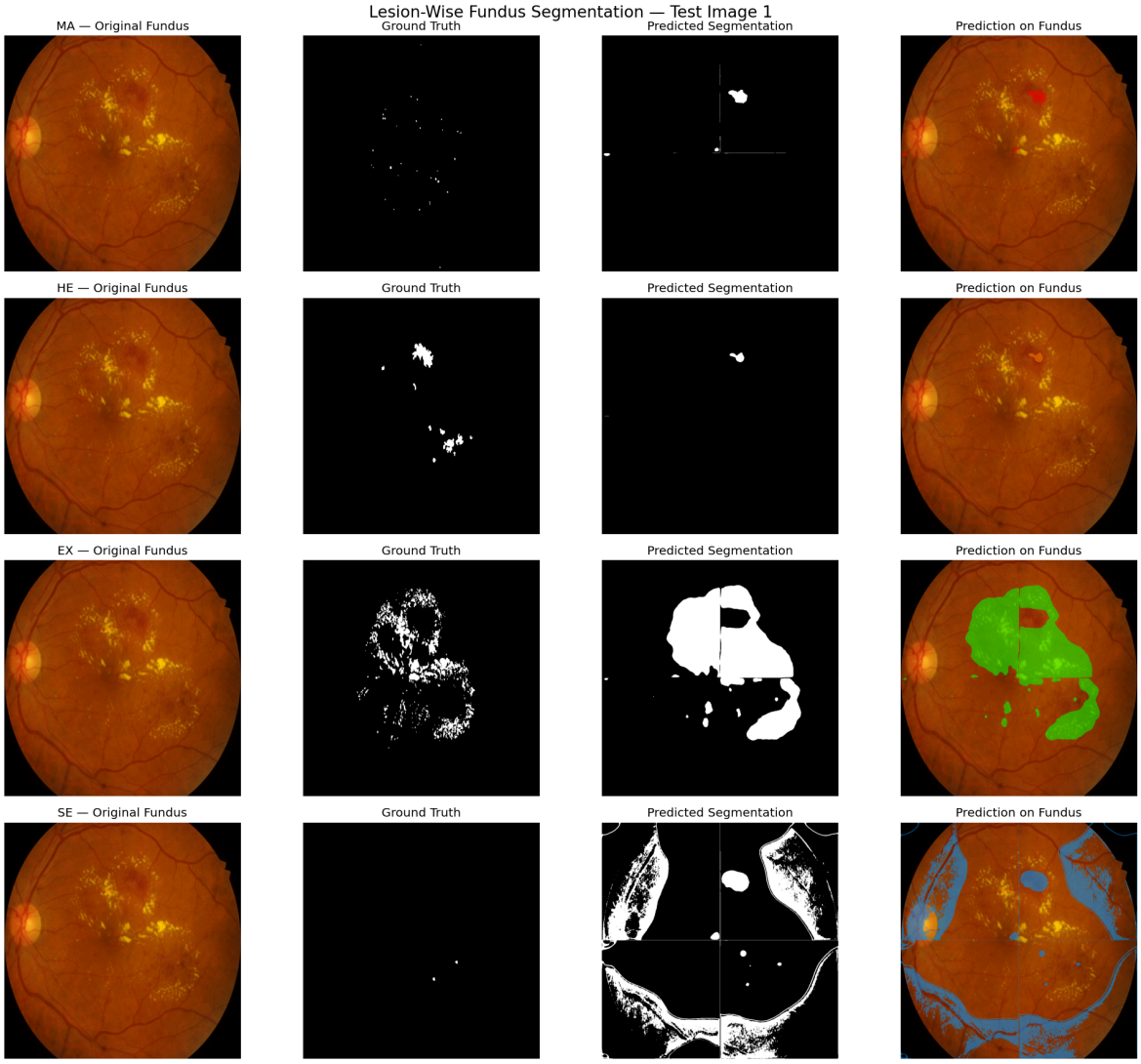
Segmentation comparison skipped because these notebook variables are unavailable: ['device']
Run the training notebook cell first, or restore the model and data-loading functions.

==========
VISUALIZATION GENERATION COMPLETED
==========
Output folder: /content/drive/MyDrive/GlucoIris/lesion_model/visualizations

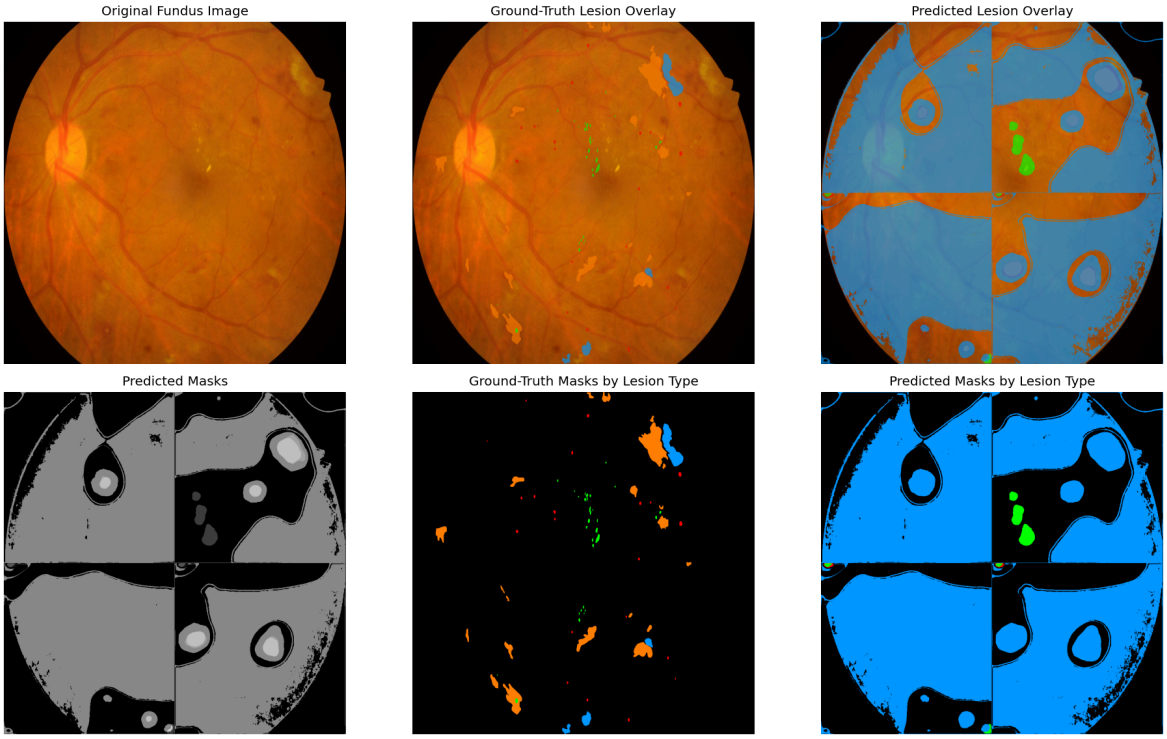
Files include:
- all_confusion_matrices.png

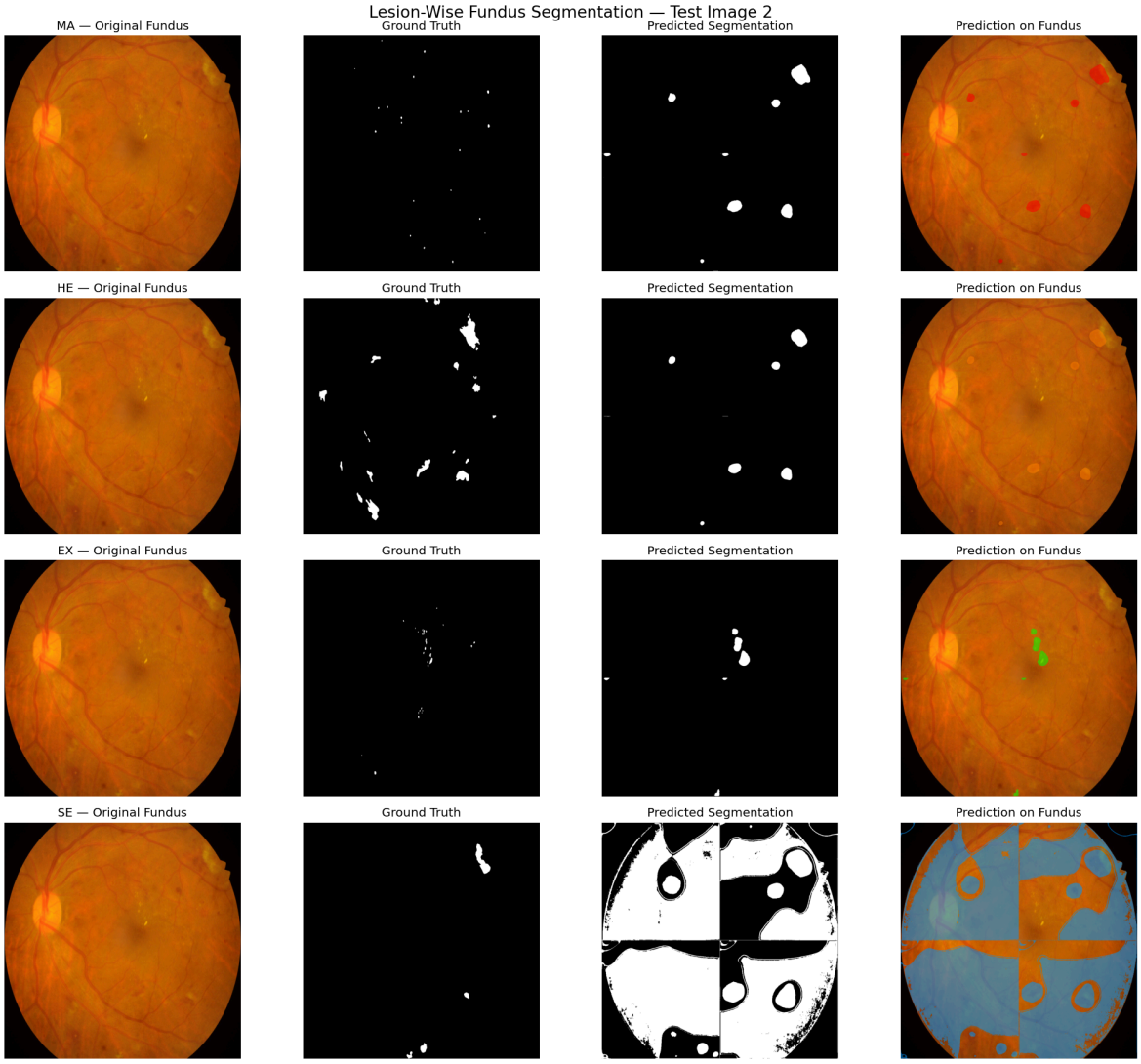
- all_metrics_comparison.png
- aupr_vs_dice.png
- metric_*.png
- training_loss_curve.png (if history format is supported)
- validation_aupr_curve.png (if history format is supported)
- before_after_all_lesions.png (if model/data variables are available)
- before_after_MA.png, before_after_HE.png, before_after_EX.png, before_after_SE.png
- lesion_metrics_summary.csv

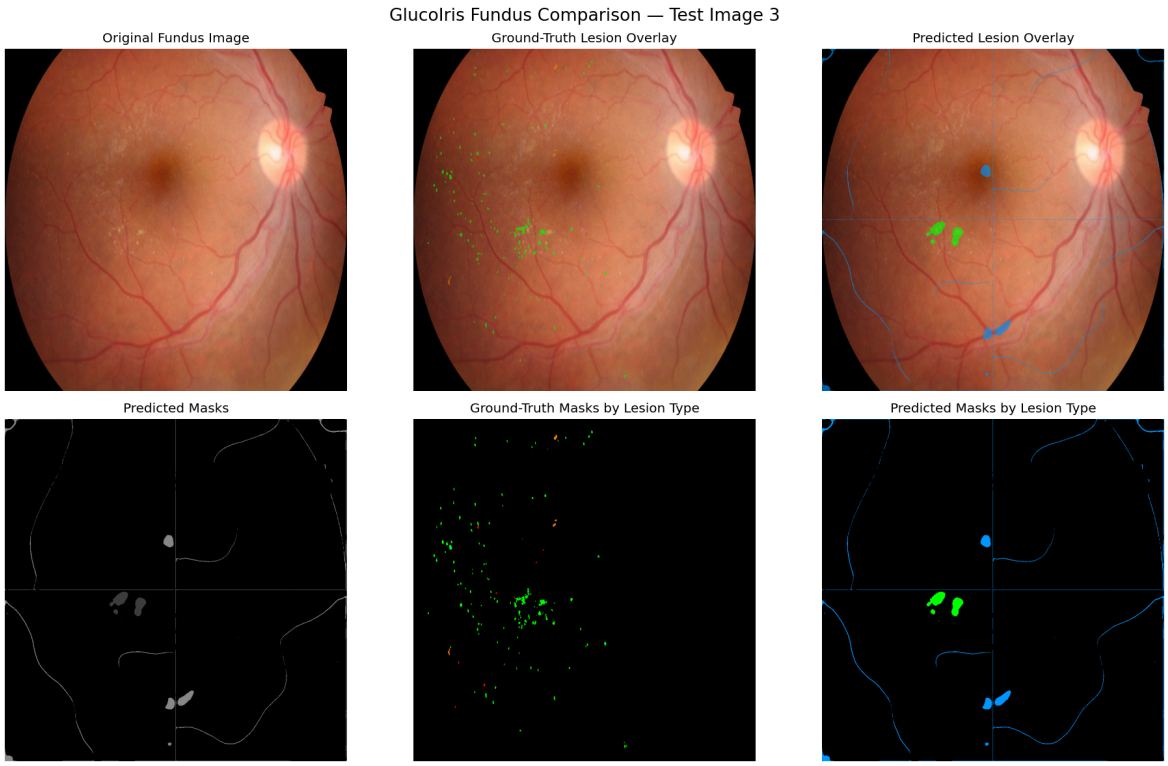


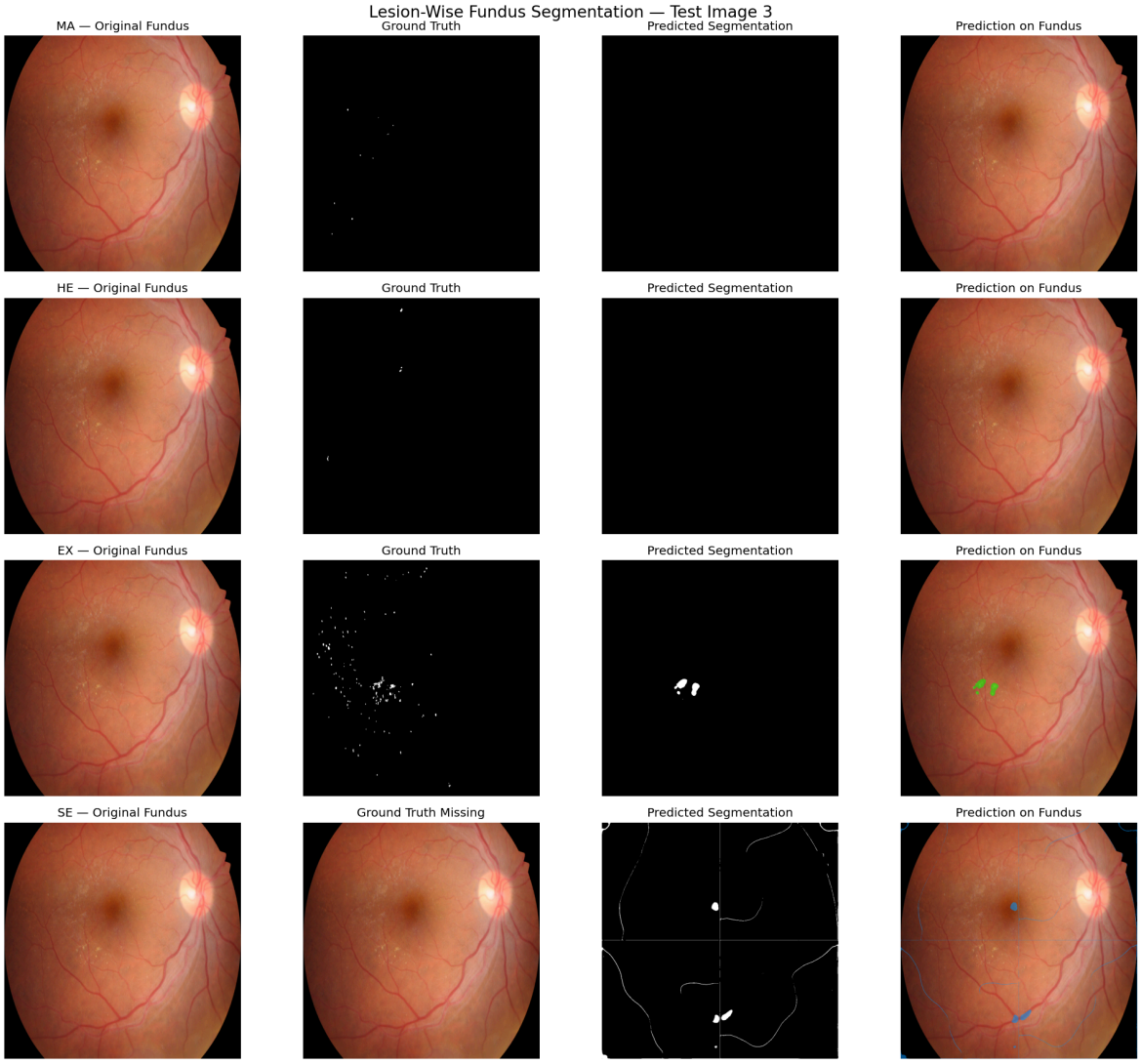

Lesion-Wise Fundus Segmentation — Test Image 1
MA — Original Fundus
Ground Truth
Predicted Segmentation
Prediction on Fundus
HE — Original Fundus
Ground Truth
Predicted Segmentation
Prediction on Fundus
EX — Original Fundus
Ground Truth
Predicted Segmentation
Prediction on Fundus
SE — Original Fundus
Ground Truth
Predicted Segmentation
Prediction on Fundus

GlucoIris Fundus Comparison — Test Image 2



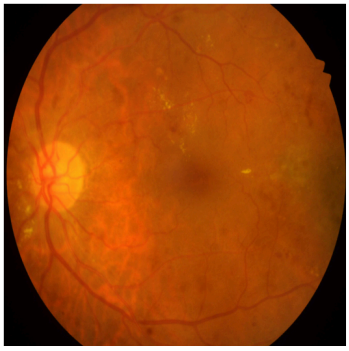

Lesion-Wise Fundus Segmentation — Test Image 2
MA — Original Fundus
Ground Truth
Predicted Segmentation
Prediction on Fundus
HE — Original Fundus
Ground Truth
Predicted Segmentation
Prediction on Fundus
EX — Original Fundus
Ground Truth
Predicted Segmentation
Prediction on Fundus
SE — Original Fundus
Ground Truth
Predicted Segmentation
Prediction on Fundus


GlucoIris Fundus Comparison — Test Image 3
Original Fundus Image
Ground-Truth Lesion Overlay
Predicted Lesion Overlay
Predicted Masks
Ground-Truth Masks by Lesion Type
Predicted Masks by Lesion Type

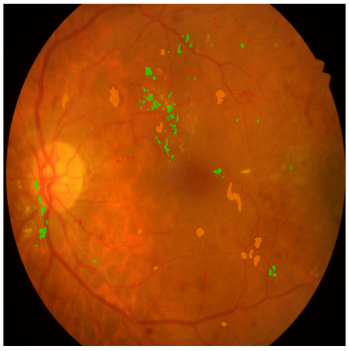

Lesion-Wise Fundus Segmentation — Test Image 3
MA — Original Fundus
Ground Truth
Predicted Segmentation
Prediction on Fundus
HE — Original Fundus
Ground Truth
Predicted Segmentation
Prediction on Fundus
EX — Original Fundus
Ground Truth
Predicted Segmentation
Prediction on Fundus
SE — Original Fundus
Ground Truth Missing
Predicted Segmentation
Prediction on Fundus

GlucoIris Fundus Comparison — Test Image 4

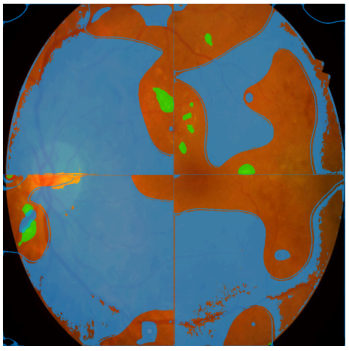
Original Fundus Image



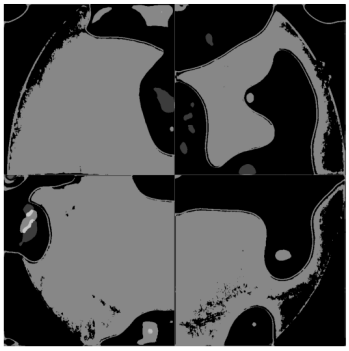
Ground-Truth Lesion Overlay



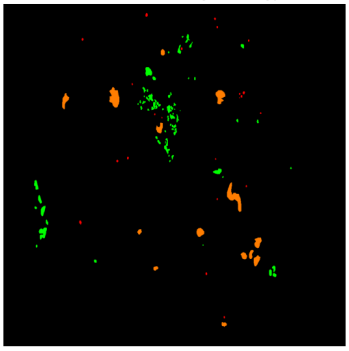
Predicted Lesion Overlay



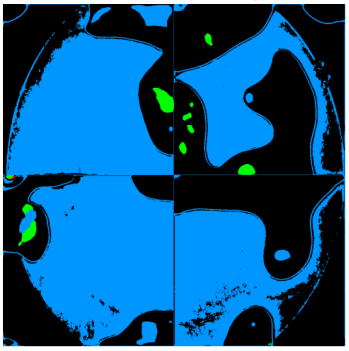
Predicted Masks

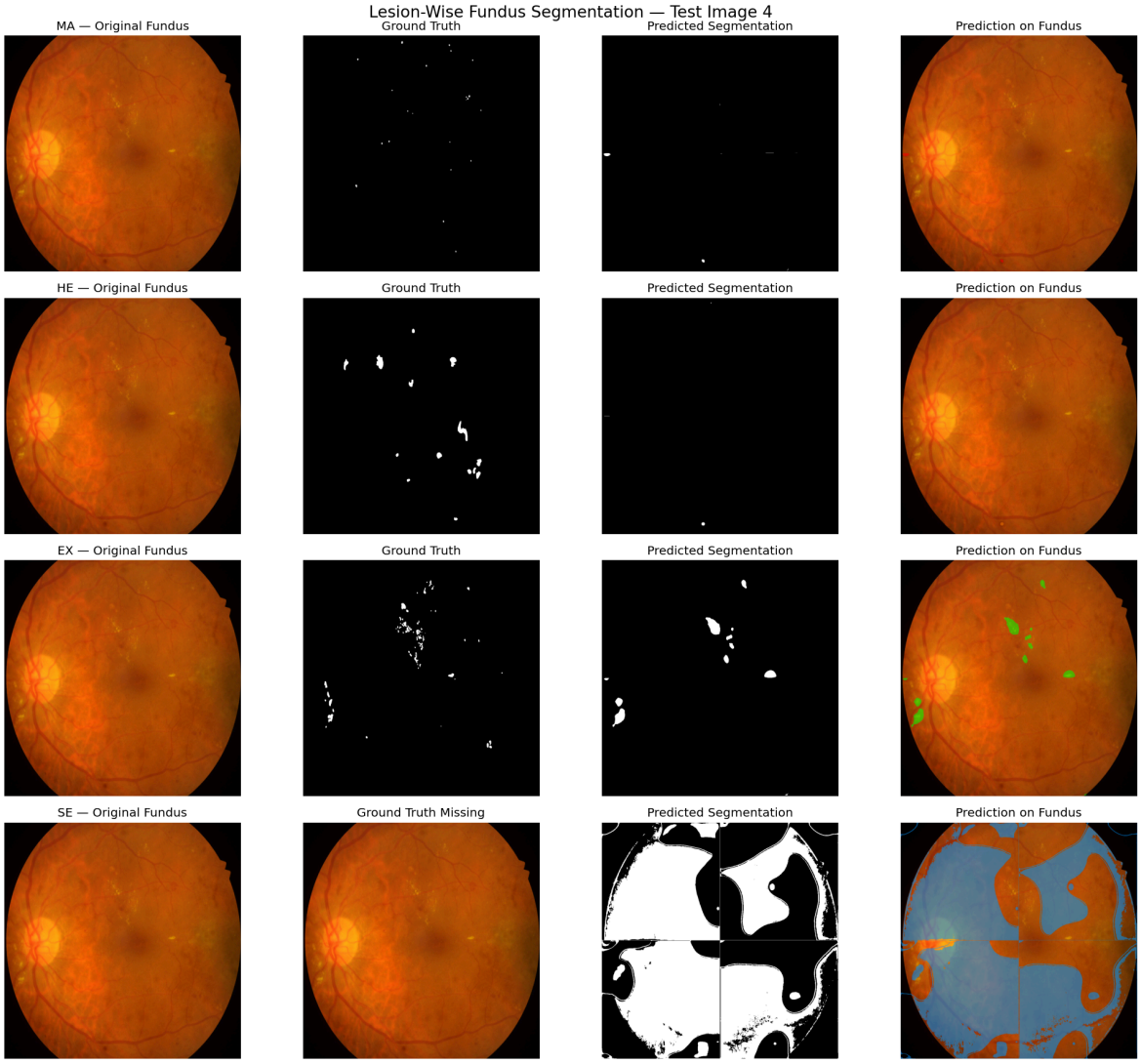


Ground-Truth Masks by Lesion Type



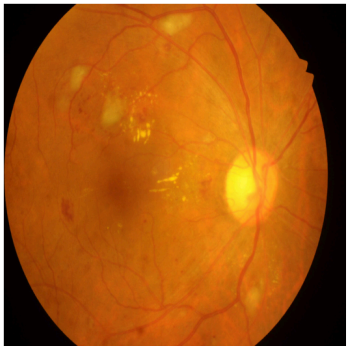
Predicted Masks by Lesion Type



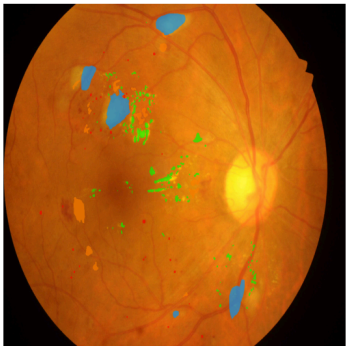

Lesion-Wise Fundus Segmentation — Test Image 4
MA — Original Fundus
Ground Truth
Predicted Segmentation
Prediction on Fundus
HE — Original Fundus
Ground Truth
Predicted Segmentation
Prediction on Fundus
EX — Original Fundus
Ground Truth
Predicted Segmentation
Prediction on Fundus
SE — Original Fundus
Ground Truth Missing
Predicted Segmentation
Prediction on Fundus

GlucoIris Fundus Comparison — Test Image 5

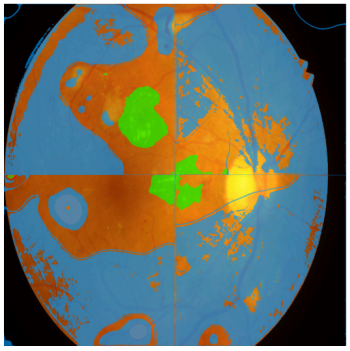
Original Fundus Image



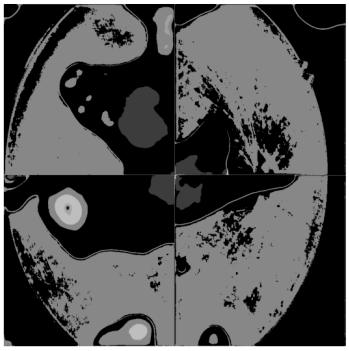
Ground-Truth Lesion Overlay



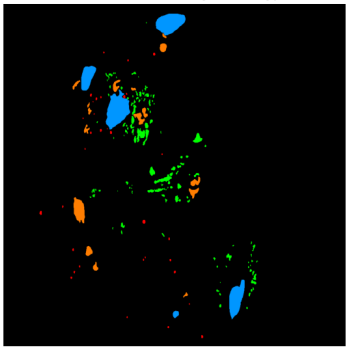
Predicted Lesion Overlay



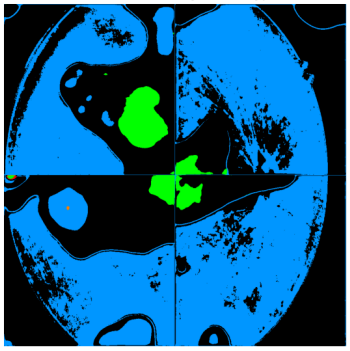
Predicted Masks

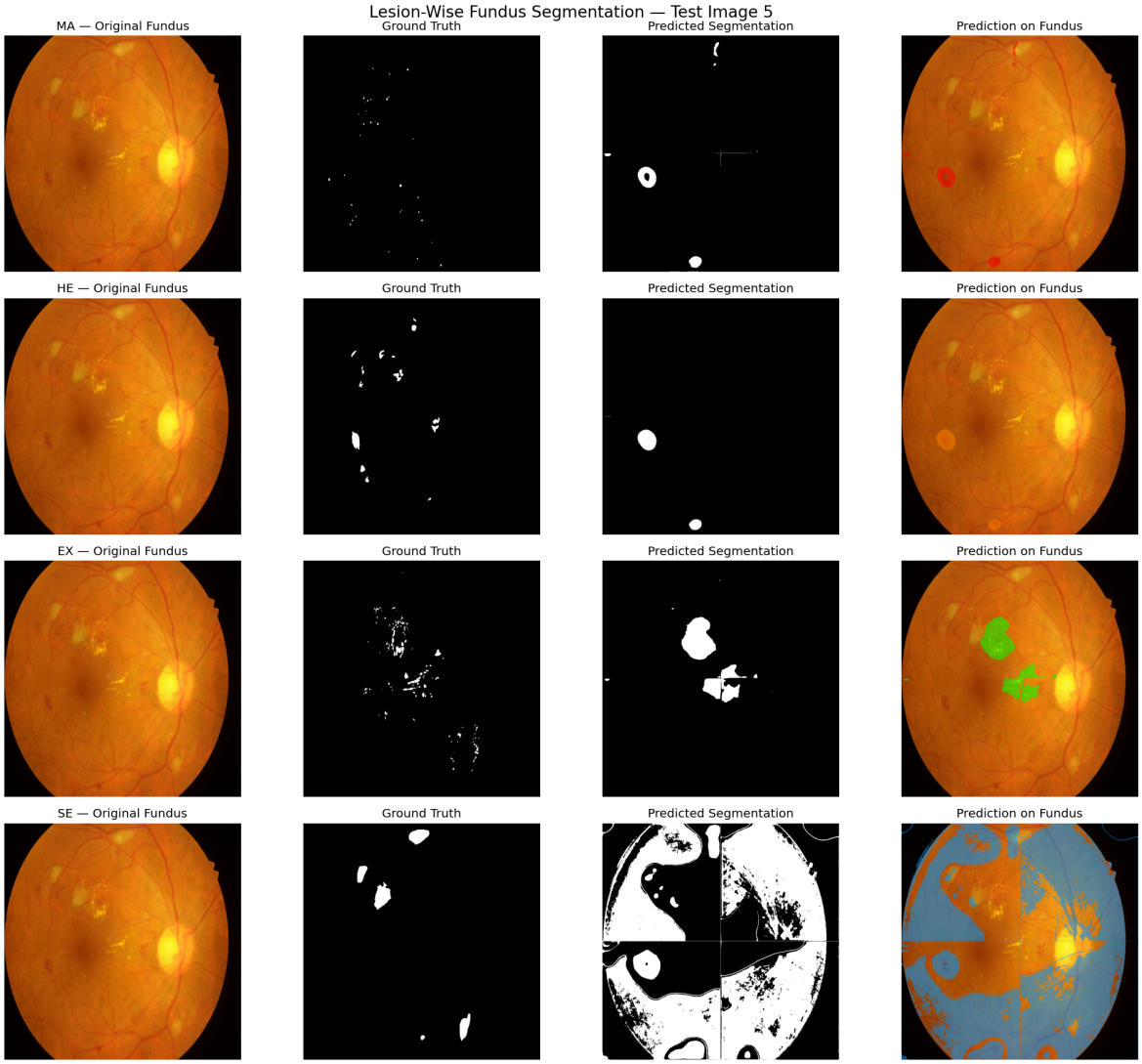


Ground-Truth Masks by Lesion Type



Predicted Masks by Lesion Type





Fundus comparisons saved to:
/content/drive/MyDrive/GlucoIris/lesion_model/visualizations/fundus_comparisons